

A Verification Pearl

The two Dempwolff–Müller polynomials are the *only* ones,
in the smallest field

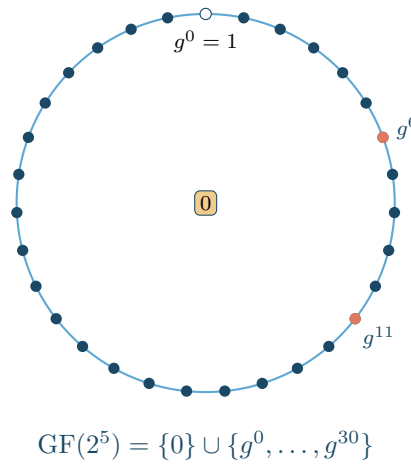
A one-page, fully machine-checked classification in Lean 4. Companion file: `pearl.lean`

The pearl. Let $L(X) = X + X^2 + X^4$ over $\text{GF}(2^5)$.
 $L(X) \cdot X^k$ is a permutation of $\text{GF}(2^5) \iff k \equiv 0, 6, \text{ or } 11 \pmod{31}$.

The paper builds permutation polynomials $L(X) \cdot X^k$ over $\text{GF}(2^n)$. For the smallest interesting case ($n = 5, m = 3$) we *classify all of them*: apart from the trivial $k = 0$, the exponents are exactly the paper's k and its companion k' . Nothing else works.

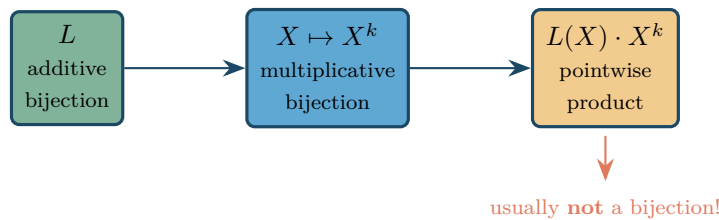
1. The field $\text{GF}(2^5)$: a clock of 31

Nonzero elements = powers of a generator g . Order 31 (prime).



Concrete model (all checked by `native_decide`). Elements = bytes 0..31. Add = bitwise xor. Multiply = carry-less product reduced mod the primitive polynomial $X^5 + X^2 + 1 = 100101_2$. Lemmas `field_comm`, `field_assoc`, `field_distrib`, `field_one`, `field_inverses`, `field_order` certify this really is $\text{GF}(2^5)$.

2. Two easy bijections, one hard product



L is a permutation; each X^k ($k = 1..30$) is a permutation. But the pointwise product of two permutations is *rarely* a permutation.

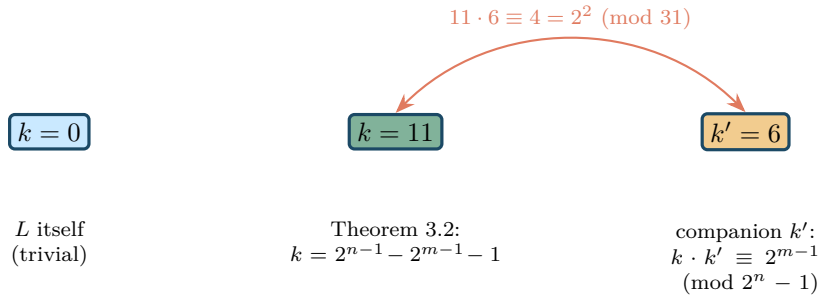
3. The classification: only 3 exponents survive

Is $L(X) \cdot X^k$ a permutation? (green = yes)

0	1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20	21
22	23	24	25	26	27	28	29	30		

Out of the 30 permutation monomials, exactly $\{6, 11\}$ keep the product a permutation. Verified exhaustively:
`dm_classification, dm_rarity.`

4. What the three winners are



So the paper's two constructions (k and k') are, for this field, the whole story.

Checked: `dm_paper_k, dm_paper_k', kk'_relation, mono11_perm.`

5. Machine-checked

Every claim above is a Lean 4 theorem in `pearl.lean`, closed by `native_decide` (allowed axioms only: `propext`, `Lean.ofReduceBool`, `Lean.trustCompiler`). No sorry.

Novelty. The concrete field, the polynomials, and the exhaustive classification are not in Mathlib; the result sharpens Theorem 3.2 by showing its two exponents are the only nontrivial ones for $\text{GF}(2^5)$.